MIT 8.012 Problem Set 4 - Problem 3.18 (K&K)

3.18 A raindrop of initial mass M_0 starts falling from rest under the influence of gravity. Assume that the drop gains mass from the cloud at a rate proportional to the product of its instantaneous mass and its instantaneous velocity:

$$\frac{dM}{dt} = kMV,$$

where k is a constant.

Show that the speed of the drop eventually becomes effectively constant, and give an expression for the terminal speed. Neglect air resistance.

Solution (handwritten) follows on the next pages.

Problem (7) (7.18 in RK)

$\frac{dm}{dt} = \rho A v$ (v = velocity)

State Analysis

time t :

$$\rho\, m(t)\, v(t) = P(t)$$

time $t+\Delta t$

$$\rho\, m(t+\Delta t)\, v(t+\Delta t) = P(t+\Delta t)$$

$$F_{ext} = \lim_{\Delta t \to 0} \frac{P(t+\Delta t) - P(t)}{\Delta t}$$

$$mg = \lim_{\Delta t \to 0} \frac{m(t+\Delta t)v(t+\Delta t) - m(t)v(t)}{\Delta t}$$

$$mg = \lim_{\Delta t \to 0} m(t)\left(\frac{v(t+\Delta t) - v(t)}{\Delta t}\right) - \frac{m(t)v(t+\Delta t) + m(t+\Delta t)v(t+\Delta t)}{\Delta t}$$

$$= \lim \frac{dv}{dt} + v(t+\Delta t)\left(\frac{m(t+\Delta t) - m(t)}{\Delta t}\right)$$

$$\lim_{\Delta t \to 0} (v(t+\Delta t)) = v(t)$$

$$m\frac{dv}{dt} + v\frac{dm}{dt} = mg$$

$$m\frac{dv}{dt} + v^2 km = mg$$

$$\frac{dv}{dt} = g - kv^2$$

$$\int \frac{dv}{g - kv^2} = t$$

$$\sqrt{g}\sin\theta = \sqrt{k}\,v \;\Big|\; g - kv^2 = g\cos^2\theta$$

$$\sqrt{g}\cos\theta\, d\theta = \sqrt{k}\, dv$$

$$\sqrt{k}\,v = \sqrt{g}$$

$$\int \sqrt{\frac{g}{k}}\,\frac{\sec\theta\tan\theta\, d\theta}{g\cos^2\theta}$$

$$= \frac{1}{\sqrt{gk}}\int \sec\theta\, d\theta = \frac{1}{\sqrt{gk}}\ln|\sec\theta + \tan\theta| + c$$

$$= \frac{1}{\sqrt{gk}}\ln\left|\frac{\sqrt{g} + \sqrt{k}\,v}{\sqrt{g - kv^2}}\right| + c$$

$$\frac{1}{\sqrt{gk}} \ln\left(\frac{\sqrt{g}+\sqrt{k}v}{\sqrt{g}-kv^2}\right) = t + C$$

$$\frac{1}{2} \ln\left(\frac{(\sqrt{g}+\sqrt{k}v)^2}{g-kv^2}\right) = C\sqrt{gk}\,t + C$$

$$\ln\left(\frac{\sqrt{g}+\sqrt{k}v}{\sqrt{g}-\sqrt{k}v}\right) = 2\sqrt{gk}\,t + C$$

$$\frac{\sqrt{g}+\sqrt{k}v}{\sqrt{g}-\sqrt{k}v} = C \cdot e^{2\sqrt{gk}\,t}$$

$$\sqrt{g} + \sqrt{k}v = \sqrt{g}\,C \cdot e^{2\sqrt{gk}\,t} - \sqrt{k}vC \cdot e^{2\sqrt{gk}\,t}$$

$$\sqrt{k}v\left(1 + C \cdot e^{2\sqrt{gk}\,t}\right) = \sqrt{g}\left(C \cdot e^{2\sqrt{gk}\,t} - 1\right)$$

$$v = \sqrt{\frac{g}{k}}\left(\frac{C \cdot e^{2\sqrt{gk}\,t} - 1}{1 + C \cdot e^{2\sqrt{gk}\,t}}\right)$$

$$= \sqrt{\frac{g}{k}}\left(1 - \frac{2}{(1 + C \cdot e^{2\sqrt{gk}\,t})}\right)$$

$\approx \sqrt{\frac{g}{k}}$ For large t.